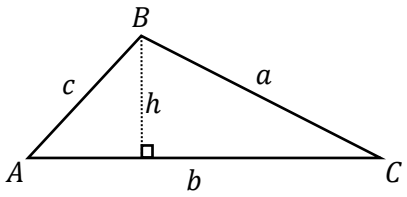
The Law of Sines

Area of a Triangle

We can develop the Law of Sines by combining two familiar topics, ***area of triangles*** and ***right triangle trigonometry***.

EXAMPLE 1

Use the familiar formula for area of a triangle, $Area = \frac{1}{2}(base)(height)$, along with right triangle trigonometry, to verify each of the area rules given below.



(a) $Area = \frac{1}{2}bc\sin(A)$

$Area = \frac{1}{2}(base)(height)$

$= \frac{1}{2}(b)(c\sin A)$

$= \frac{1}{2}bc\sin A$

$\sin A = \frac{h}{c}$

$h = c\sin A$

(b) $Area = \frac{1}{2}ab\sin(C)$

$Area = \frac{1}{2}(base)(height)$

$= \frac{1}{2}(b)(a\sin C)$

$= \frac{1}{2}ab\sin C$

$\sin C = \frac{h}{a}$

$h = a\sin C$

THEOREM: *Area of a Triangle*

For any triangle ABC, with side measures a, b, and c, the area is given by any of the following formulas.

$$Area = \frac{1}{2}bc\sin(A) = \frac{1}{2}ac\sin(B) = \frac{1}{2}ab\sin(C)$$

While the above result provides a useful means for calculating the area of a triangle, perhaps more importantly, it establishes a valuable equivalence relationship. The Law of Sines is a direct consequence of the above statement.

Applying the Law of Sines

THEOREM: *Law of Sines*

For any triangle ABC, with side measures a, b, and c, we have the following relationship.

$$\frac{\sin(A)}{a} = \frac{\sin(B)}{b} = \frac{\sin(C)}{c}$$

Alternatively, we can restate the above as follows:

$$\frac{a}{\sin(A)} = \frac{b}{\sin(B)} = \frac{c}{\sin(C)}$$

EXAMPLE 2

Show that the law of sines is a direct consequence of the area formulas established earlier.

$2\left(\frac{1}{2}bc\sin A\right) = 2\left(\frac{1}{2}ac\sin B\right) = 2\left(\frac{1}{2}ab\sin C\right)$

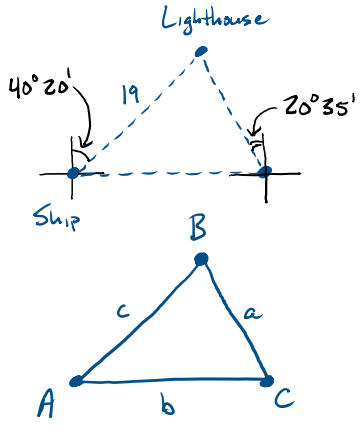
$\frac{bc\sin A}{abc} = \frac{ac\sin B}{abc} = \frac{ab\sin C}{abc}$

$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

Equipped with the law of sines, we now have a powerful tool that can be useful in a variety of applications.

EXAMPLE 3

A ship sailing due east spots a lighthouse 19 km away located at a bearing of $N\ 40°20'\ E$. Thirty minutes later, the lighthouse is located at a bearing of $N\ 20°35'\ W$. How fast is the ship sailing in km/hr?



We know

$A = 49°40'$

$C = 69°25'$

$c = 19$

$*B = 60°55'$

↑ sum = 180°

$49°40' + 69°25' = 119°5'$

$179°60' - 119°5' = 60°55'$

We need

$b = ?$

$\frac{b}{\sin B} = \frac{c}{\sin C}$

$b = \sin B\left(\frac{c}{\sin C}\right)$

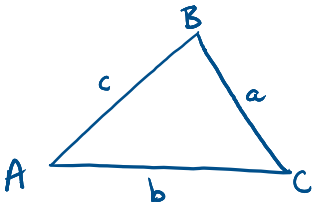
$b = \sin\left(60+\frac{55}{60}\right)\left(\frac{19}{\sin\left(69+\frac{25}{60}\right)}\right)$

$b \approx (17.73663345\ldots) \times 2 \to 35.47$

Sailing at approx $35.47\frac{km}{hr}$

EXAMPLE 4

Solve the triangle ABC given $A = 32.2°$, $B = 52.4°$ and $b = 42$ cm.



We know
$A = 32.2$
$B = 52.4$
$b = 42$
$C = 95.4$ (sum = 180°)

We need
$c = ?$
$a = ?$

Find c

$\frac{c}{\sin C} = \frac{b}{\sin B}$

$c = \sin(95.4) \cdot \frac{42}{\sin(52.4)}$

$c \approx 52.78$ cm

Find a

$\frac{a}{\sin A} = \frac{b}{\sin B}$

$a = \sin(32.2) \cdot \frac{42}{\sin(52.4)}$

$a \approx 28.25$ cm

The Ambiguous Case of the Law of Sines

You may have noticed in each of the preceding examples we were provided with ***two angle measures and one side measure***. With this information, we can quickly determine the third missing angle and apply the law of sines to solve for all remaining side measures. However, we must exercise caution when only one angle measure is provided. In particular, with only one angle and two side measures (one of which must be opposite the given angle to apply the law of sines), we may arrive at one of three possible outcomes, (1) a single unique triangle exists, (2) two triangles exist, and (3) no such triangle exists. Let's review an example for each case.

EXAMPLE 5

Solve the triangle ABC given $a = 52.4$ cm, b= 32.2 cm, and $B = 95°$.

We know
$a = 52.4$
$b = 32.2$
$B = 95$

We need
$A =$
$C =$
$c =$

Find A

$\frac{\sin A}{a} = \frac{\sin B}{b}$

$\sin A = \frac{a \sin B}{b}$

$A = \sin^{-1}\left(\frac{52.4 \sin(95)}{32.2}\right)$ → Calc. gives ERROR message!

$A = \sin^{-1}(1.62)$ not possible

No triangle exists!

In the last example, we found that no such triangle exists satisfying the given information. In fact, we can apply the following theorem to establish the same result.

THEOREM: *The Longest Side Theorem*

If one side of a triangle is longer than another side, then the angle opposite the longer side is the larger angle.

Conversely, if one angle of a triangle is larger than another angle, then the side opposite the larger angle is the longer side.

On your own, apply the above theorem to our last example to verify our conclusion that no such triangle exists satisfying the given information.

EXAMPLE 6

Solve the triangle ABC given $a = 18.2$ cm, b= 22.5 cm, and $B = 65°$.

We know

$a = 18.2$

$b = 22.5$

$B = 65°$

We need

$A =$

$C =$

$c =$

Find A

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\sin A = \frac{a \sin B}{b}$$

$$A = \sin^{-1}\left(\frac{18.2 \sin(65)}{22.5}\right)$$

$A_1 \approx 47.15°$

↑ ONE of TWO possibilities!

Recall $\sin(\theta°) = \sin(180° - \theta°)$

$A_2 = 180° - 47.15° = 132.85°$

Triangle 1 $(A_1 \approx 47.15°)$

$C_1 = 180 - B - A_1$

$C_1 \approx 180 - 65 - 47.15$

$C_1 \approx 67.85°$

$$\frac{c_1}{\sin C_1} = \frac{b}{\sin B}$$

$$c_1 \approx \sin(67.85) \cdot \frac{22.5}{\sin(65)}$$

$c_1 \approx 22.99 \rightarrow 23$ cm

Triangle 2 $(A_2 \approx 132.85°)$

$C_2 = 180 - B - A_2$

$C_2 \approx 180 - 65 - 132.85$

$C_2 \approx -17.85°$ not possible

Only one triangle

EXAMPLE 7

Solve the triangle ABC given $c = 32.5$ cm, $b = 29.7$ cm, and $B = 65°$.

We know

$b = 29.7$

$c = 32.5$

$B = 65$

We need

$A =$

$C =$

$a =$

Find C

$$\frac{\sin C}{c} = \frac{\sin B}{b}$$

$$\sin C = \frac{c \sin B}{b}$$

$$C = \sin^{-1}\left(\frac{32.5 \sin(65)}{29.7}\right)$$

$C_1 \approx 82.64$

↑ ONE of TWO possibilities! $C_2 = 180 - 82.64 = 97.36°$

Triangle 1 ($C_1 \approx 82.64°$)

$A_1 = 180 - B - C_1$

$A_1 \approx 180 - 65 - 82.64$

$A_1 \approx 32.36°$

$$\frac{a_1}{\sin A_1} = \frac{b}{\sin B}$$

$$a_1 \approx \sin(32.36) \cdot \frac{29.7}{\sin(65)}$$

$a_1 \approx 17.54 \rightarrow 17.54$ cm

Triangle 2 ($C_2 \approx 97.36°$)

$A_2 = 180 - B - C_2$

$A_2 \approx 180 - 65 - 97.36°$

$A_2 \approx 17.64°$

$$\frac{a_2}{\sin A_2} = \frac{b}{\sin B}$$

$$a_2 \approx \sin(17.64) \cdot \frac{29.7}{\sin(65)}$$

$a_2 \approx 9.93 \rightarrow 9.93$ cm

Two possible triangles!